

# Incident Ticket #1: Inter-VLAN Reachability Failure (RPC1 to Admin PC)

**Status:** Resolved

**Skill Evaluation:** Successful resolution; documented lessons learned and areas for improvement.

## Ticket Overview

- **Ticket ID:** TCK-001
- **Source:** RPC1 | **VLAN:** 10 (Reception\_Data) | **Source IP:** 192.168.10.x | **Gateway:** 192.168.10.1
- **Destination:** Admin PC | **VLAN:** 20 (Admin\_Data) | **Target IP:** 192.168.20.10 | **Gateway:** 192.168.20.1
- **Symptom:** RPC1 cannot reach Admin PC over ICMP.
- **Expected Behavior:** 0% packet loss across VLANs via Router-on-a-Stick (ROAS).
- **Actual Behavior:** 100% packet loss (Request Timed Out).

## Root Cause Analysis

1. **Switch B Trunk Failure:** Interface Fa0/2 on Switch B was disabled, preventing traffic flow to the Admin VLAN.
2. **Host IP Configuration Error:** RPC1 was configured with an invalid IP address and incorrect Default Gateway.
3. **VLAN Assignment Error:** RPC1 access port on Switch A was assigned to VLAN 20 instead of VLAN 10 during investigation.

## Step-by-Step Resolution

[RPC1] ---> (SW A Fa0/4) ---> [Core SW] ---> (Fa0/2 Down) -X-> [SW B] ---> [Admin PC]

|  
(Fixed Trunk & Gateway)  
v

[RPC1] -----> [Core SW] -----> [SW B] ---> [Admin PC]

1. **Link Restoration:** Re-enabled uplink interface **Fa0/2** on Switch B and explicitly set trunking mode:

```
Switch-B(config)# interface FastEthernet0/2
Switch-B(config-if)# switchport mode trunk
Switch-B(config-if)# no shutdown
```

2. **Access VLAN Correction:** Reassigned Switch A port Fa0/4 to Reception\_Data (VLAN 10).
3. **Host Addressing Update:** Released/renewed DHCP settings on RPC1 to pull the proper default gateway (192.168.10.1).

## Verification Proof

```
C:\> ping 192.168.20.10
```

Pinging 192.168.20.10 with 32 bytes of data:

```
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
```

```
Reply from 192.168.20.10: bytes=32 time=19ms TTL=127
```

```
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
```

```
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
```

Ping statistics for 192.168.20.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)

## Key Technical Takeaways

- Verify target IP addresses early in the process to prevent testing invalid network paths.
- Use path-based troubleshooting: verify source host configuration, local gateway reachability, core transit paths, destination gateway, and finally the target host